

Notes: Frydman & Saks (RFS, 2010)

Executive Compensation: A New View from a Long-Term Perspective, 1936–2005

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1 Big picture

- **Question.** How did the level, structure, and incentive strength of executive compensation evolve over the long run, from 1936 to 2005?
- **Main data contribution.** The paper hand-collects a long-run panel of the three highest-paid executives in large U.S. publicly traded firms using proxy statements and 10-K reports.
- **Main level-of-pay fact.** Median real executive compensation was roughly flat from the late 1940s through the mid-1970s, even though firms grew substantially.
- **Modern contrast.** After the mid-1970s, and especially after the 1980s, executive pay rose rapidly and became much more correlated with aggregate firm size and stock market growth.
- **Structure of pay.** Before 1980, compensation was mainly salary and current bonus. Stock options and long-term incentives became important over time, especially from the 1980s onward.
- **Tax result.** Pre-tax and after-tax compensation follow similar long-run patterns. The rise in pay is not simply an artifact of changing tax rates.
- **CEO versus other executives.** The CEO premium is not a recent invention, but it rises sharply in recent decades. The CEO-to-other-top-officer pay ratio is about 2.58 in 2000–2005.
- **Firm-size relationship.** The strong relationship between executive pay and firm size in recent decades was not present for most of the twentieth century. This challenges explanations that rely only on growth in aggregate firm size.
- **Pay-to-performance fact.** Executive wealth was sensitive to firm performance for most of the sample, not only in recent decades. Incentives weakened in the 1970s but strengthened sharply after the mid-1980s.
- **Challenge to governance/rent explanations.** Pay and option use were lower from the 1950s to 1970s even though corporate governance was not obviously stronger then.
- **Challenge to pure market-size explanations.** If larger firm size mechanically raised compensation, pay should have grown much more in the postwar decades before 1975.
- **Takeaway.** The long-run data show that the recent explosion of executive pay is historically unusual and cannot be explained by one simple mechanism such as firm size, options, taxes, or weak governance alone.

2 Data and measurement

2.1 Sample construction

- The sample is based on the three highest-paid officers in the largest fifty firms in 1940, 1960, and 1990.
- Compensation data are collected from proxy statements and 10-K reports.
- The sample covers 1936–2005.
- The final dataset contains 15,883 person-year observations and 2,862 individual executives.
- Firms are large relative to the economy: the average number of firms observed in each year is 76.
- The median executive is observed for four years, and the average executive is observed for 5.6 years.

Interpretation: The sample is not a random sample of all executives. It is designed to describe high-level executives in large publicly traded firms, the group most relevant for debates about top executive compensation.

2.2 Total compensation

The paper defines total compensation as

$$TotalComp = Salary + CurrentBonus + LongTermBonus + BSV\text{value}(OptionGrants). \quad (1)$$

- Salary and current bonus are the cash components of pay.
- Long-term bonuses include deferred bonus and long-term incentive payments.
- Option grants are valued using a Black–Scholes method.
- Real values are measured in year-2000 dollars.

Interpretation: The key empirical contribution is consistency. Earlier historical studies valued different components in inconsistent ways, often overstating mid-century option compensation.

2.3 Compensation structure

The paper separates compensation into three cumulative measures:

1. salary plus current bonus;
2. salary plus current bonus plus long-term pay;
3. salary plus current bonus plus long-term pay plus option grants.

Interpretation: This decomposition shows which components caused the rise in compensation. The large modern increase is strongly connected to long-term pay and stock options.

2.4 Firm size and aggregate firm growth

Firm size is measured mainly as market value, and robustness checks also use sales. The core long-run comparison is between

$$\ln(TotalComp_{it}) \quad \text{and} \quad \ln(FirmSize_{it}). \quad (2)$$

Interpretation: The paper tests whether the increase in pay can be explained by the growth of firms. The answer is partly yes in recent decades, but mostly no for the postwar decades before the mid-1970s.

2.5 Stock and option holdings

The paper observes executives' holdings of stock and stock options over the entire sample period. This permits a long-run measure of the sensitivity of executive wealth to firm performance.

Two main measures are used:

$$JM_{it} = 1000 \times \frac{\Delta W_{it}}{\Delta FirmValue_{it}}, \quad (3)$$

where JM is the Jensen–Murphy statistic: the dollar change in executive wealth for a \$1,000 change in firm market value.

$$EquityStake_{it} = 0.01 \times FirmValue_{it} \times \frac{\Delta W_{it}}{\Delta FirmValue_{it}}, \quad (4)$$

which is the dollar change in executive wealth for a 1% increase in the firm's rate of return.

Interpretation: The Jensen–Murphy statistic scales by firm market value, while equity-at-stake measures the actual dollar gain from a percentage improvement in firm performance.

2.6 After-tax compensation

The paper also constructs after-tax compensation using historical tax rules. The after-tax series assumes the executive files jointly with a spouse, takes no deductions, and has no other income.

Interpretation: Comparing pre-tax and after-tax compensation asks whether the long-run pattern is driven by tax policy. The figures show that taxes do not overturn the main trend.

3 Models and empirical specifications

3.1 Trend decomposition

The descriptive analysis tracks the median of compensation and its components over time:

$$MedianComp_t = median_i(TotalComp_{it}). \quad (5)$$

The paper also compares alternative compensation series:

$$Comp_t^{Cash}, \quad Comp_t^{LT}, \quad Comp_t^{Options}. \quad (6)$$

Interpretation: The goal is not only to document the level of pay, but to show which component of pay changed over time.

3.2 Firm-size relationship

The paper studies regressions of the form

$$\ln(Comp_{i,f,t+1}) = \alpha + \beta_1 AverageSize_{t-1} + \beta_2 RelativeSize_{i,f,t-1} + \epsilon_{i,f,t}. \quad (7)$$

Some specifications include firm fixed effects:

$$\ln(Comp_{i,f,t+1}) = \alpha_f + \beta_1 AverageSize_{t-1} + \beta_2 RelativeSize_{i,f,t-1} + \epsilon_{i,f,t}. \quad (8)$$

- *AverageSize* captures aggregate market size.
- *RelativeSize* captures firm-specific size relative to market and year averages.
- Separate estimates are reported for 1936–1975, 1946–1975, and 1976–2005.

Interpretation: This separates market-wide firm growth from firm-specific variation. The central finding is that average firm size explains much more compensation variation after 1976 than before 1975.

3.3 Pay-to-performance measurement

The ex ante revaluation of executive wealth from stock and option holdings can be summarized as

$$\Delta W_{it} = Shares_{it} \Delta P_{ft} + \sum_j OptionDelta_{ijt} \Delta P_{ft}. \quad (9)$$

The paper reports the resulting dollar change in wealth for:

- a \$1,000 increase in firm market value;
- a 1% increase in the firm's rate of return.

Interpretation: By measuring both stock and option holdings, the paper avoids the mistake of studying only salaries and bonuses when assessing incentives.

3.4 Determinants of pay-to-performance

The paper studies the cross-sectional relationship between pay-to-performance and firm characteristics:

$$P2P_{i,f,t} = \alpha + \beta' X_{f,t-1} + IndustryFE + \epsilon_{i,f,t}, \quad (10)$$

where X includes board structure, firm size, leverage, growth opportunities, idiosyncratic volatility, and real returns.

Interpretation: The determinants of incentives are not radically different across the twentieth century. This is important because it suggests that the modern incentive-pay regime did not emerge from entirely new primitives.

3.5 Strength of incentives

To quantify incentives, the paper asks how executive wealth changes when firm performance moves from the median return to the seventieth percentile return:

$$\Delta W_{it}^{70-50} = W_{it}(r^{70}) - W_{it}(r^{50}). \quad (11)$$

It also reports a compensation elasticity:

$$Elasticity = \frac{\% \Delta Wealth}{r^{70} - r^{50}}. \quad (12)$$

Interpretation: This calculation translates abstract pay-to-performance measures into a more intuitive question: how much money does an executive gain from a meaningful improvement in firm performance?

4 Identification and empirical design

4.1 What the paper identifies

- The paper does not identify a single causal shock to executive pay.
- Instead, it creates a consistent long-run time series and tests whether well-known modern explanations fit the full twentieth-century evidence.
- The identification comes from historical contrast: postwar stability in pay versus post-1980 acceleration.

Interpretation: The main contribution is measurement and long-run comparison. A theory that explains the recent data but fails in the earlier decades is incomplete.

4.2 Why the long-run perspective matters

- Post-1970 data alone make pay and firm size look tightly linked.
- The long-run data show that this link was weak from the late 1940s to the mid-1970s.
- Therefore, recent correlations may reflect a historically specific regime rather than a universal law of compensation.

Interpretation: The paper's core empirical strategy is to test modern theories against a much longer history than the usual executive-pay datasets.

4.3 Implications for major explanations

- **Rent extraction/governance.** The data do not support a simple story that pay rose only because governance weakened. Governance was not obviously stronger in the earlier low-pay decades.
- **Firm size and talent markets.** Firm size explains the post-1980 rise better than the postwar flat-pay period.
- **Risk compensation.** The increased use of incentive pay does not by itself explain the full rise in pay under stable unobserved parameters.
- **General managerial skills.** The move toward more transferable managerial talent may help explain why pay became more market-wide and less firm-specific in recent decades.

Interpretation: The paper does not reject every theory outright. It shows that each theory needs to explain why the relevant mechanism became much stronger after the 1970s.

5 Tables and figures

The visuals below are directly extracted from the paper and grouped compactly to improve reading speed.

5.1 Tables 1–2: sample and job titles

Read:

- Table 1 reports 15,883 executive-year observations and 2,862 individual executives.
- The average number of firms in each year is 76.
- Executives are mostly observed in very large firms, but the sample covers firms ranked outside the top 200 as well.
- Table 2 shows that the sample includes CEOs, presidents, chairmen, vice presidents, and other top officers.
- More than 80% of executives are also directors.

Economic message: The sample captures top decision-makers in large public firms and is broad enough to study long-run trends beyond a single firm or industry.

Table 1
Sample summary statistics

	1936–2005
Total # of person-year observations	15,883
Total # of executives	2,862
Average # of firms in each year	76
Average # of years each executive is observed	5.6
Median # of years each executive is observed	4
Fraction of obs. in firms with market value	
Ranked 1–50	39.0
Ranked 50–100	19.6
Ranked 100–200	19.1
Ranked 200–500	16.7
Ranked 500+	5.4

Based on the three highest-paid officers in the largest fifty firms in 1940, 1960, and 1990 (a total of 101 firms). Rankings by market value are based on all firms appearing in the CRSP database, which includes all publicly traded firms in the NYSE, AMEX, and NASDAQ stock exchanges. Annual market value is measured at the end of the fiscal year.

Table 1: sample summary statistics

Table 2
Distribution of job titles

	Percent of observations		
	Entire sample	1936–1969	1970–2005
Chairman of the board	21.2	15.8	25.9
Vice-chairman	6.4	2.0	10.3
President	28.5	31.6	25.9
Chief executive officer	15.3	2.3	26.8
Chief financial officer	1.8	0.0	3.4
Chief operating officer	5.0	0.2	9.1
Executive or senior vice-president	21.6	15.3	27.2
Vice-president	15.2	27.8	4.1
Treasurer	1.2	2.4	0.1
Comptroller	0.6	1.3	0.1
Other job title	8.7	8.4	9.0
Director	84.7	91.7	78.6

Based on the three highest-paid officers in the largest fifty firms in 1940, 1960, and 1990 (a total of 101 firms). The sum of each column is greater than 100% because some officers hold multiple titles. Other categories not listed include “secretary,” “chairman of the executive committee,” and officers of subsidiaries. The row labeled “director” is the percentage of executives in the sample that are also members of the board of directors.

Table 2: distribution of job titles

5.2 Figures 1–2: compensation components and stock options

Read:

- Figure 1 shows a J-shaped pattern: total pay falls during the war and early postwar decades, remains flat from the 1950s to the 1970s, and rises sharply after the 1980s.

- Salary and current bonus grow less than total compensation after the 1980s.
- Long-term pay and options account for much of the modern increase.
- Figure 2 shows that option holdings become common after the 1950s and option grants become especially frequent after the 1980s.

Economic message: The rise in executive pay is not simply a cash-pay phenomenon. It is tied to a structural change in compensation packages toward long-term and equity-linked pay.

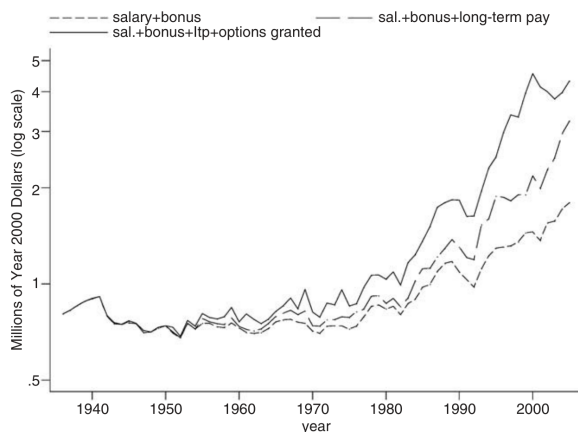


Figure 1
Median total compensation and its components, 1936–2005
Each line shows the median value of compensation defined as an increasing number of types: salary and current bonuses (paid out in stock or in cash); salary, current bonuses, and long-term incentive payments (paid out in stock or in cash); and salary, current and long-term bonuses, and the Black–Scholes value of stock options granted. Based on the three highest-paid officers in the largest fifty firms in 1940, 1960, and 1990 (a total of 101 firms).

Figure 1: median total compensation and components

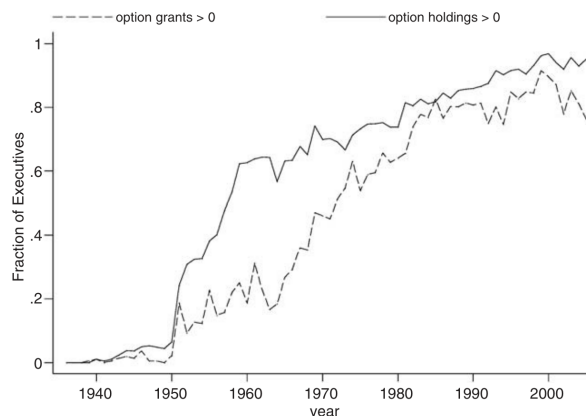


Figure 2
Fraction of top executives granted and holding stock options
Based on the three highest-paid officers in the largest fifty firms in 1940, 1960, and 1990 (a total of 101 firms). Option grants > 0 shows the fraction of executives in the year that were granted at least one stock option. Option holdings > 0 shows the fraction of executives in the year that held at least one stock option.

Figure 2: option grants and option holdings

5.3 Figure 3 and Table 3: taxes, CEO status, and percentile patterns

Read:

- Figure 3 shows that pre-tax and after-tax compensation follow similar long-run patterns.
- Excluding options also leaves the main J-shaped trend in place, although options amplify the recent increase.
- Table 3 shows that compensation increases across all percentiles after 1980, especially at the top.
- The CEO-to-other-top-officer ratio rises from about 1.5 in the prewar era to 2.58 in 2000–2005.

Economic message: Taxes do not explain the central trend. The modern increase is broad across the top executive distribution, but especially strong for CEOs and high percentiles.

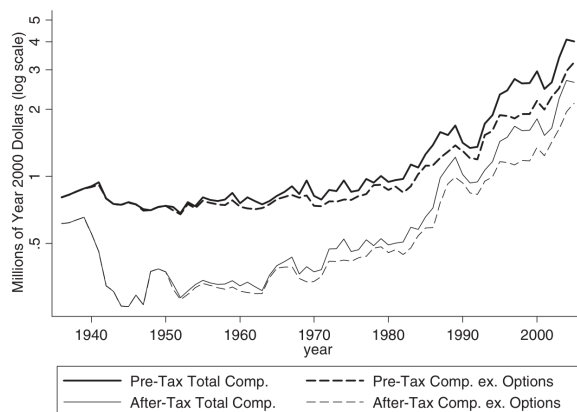


Figure 3
Pre- and after-tax compensation
Based on the three highest-paid officers in the largest fifty firms in 1940, 1960, and 1990 (a total of 101 firms). Total compensation is composed of salary, bonuses, long-term bonus payments, and stock options. Dashed lines show the median pre- and after-tax compensation excluding stock options. Options values are related to how they are taxed: we use the Black-Scholes value in the year granted from 1950 to 1977 and the gains from exercising in all other years. Options are taxed as capital gains from 1950 to 1977 and as labor income in other years. After-tax pay is calculated assuming that the executive files jointly with a spouse, takes no deductions, and has no other source of income.

Figure 3: pre- and after-tax compensation

Table 3
Level and structure of total compensation by percentile and CEO status

	1936-1939	1940-1945	1946-1949	1950-1959	1960-1969	1970-1979	1980-1989	1990-1999	2000-2005
Panel A: Level of total compensation (millions of year 2000 dollars)									
Three highest-paid officers									
10th percentile	0.36	0.40	0.36	0.39	0.45	0.47	0.57	0.91	1.31
25th percentile	0.53	0.59	0.53	0.55	0.60	0.64	0.85	1.35	2.19
50th percentile	0.85	0.80	0.72	0.77	0.83	0.93	1.33	2.36	4.08
75th percentile	1.24	1.15	1.01	1.09	1.18	1.31	2.05	4.43	9.42
90th percentile	1.80	1.59	1.53	1.63	1.66	1.84	3.18	8.29	16.9
Average	0.97	0.95	0.85	0.94	0.99	1.09	1.74	4.35	7.63
Median CEO	1.11	1.07	0.90	0.97	0.99	1.17	1.81	4.09	9.20
Median other top officers	0.74	0.70	0.65	0.67	0.74	0.82	1.12	1.89	3.02
Within-firm ratio of CEO to other top officers	1.50	1.48	1.38	1.43	1.29	1.42	1.58	2.00	2.58
Panel B: Structure of compensation									
Average long-term pay/bonus compensation									
CEOs	0.00	0.01	0.01	0.03	0.06	0.05	0.07	0.15	0.23
Other top officers	0.00	0.01	0.01	0.03	0.05	0.05	0.07	0.15	0.22
CEOs	0.00	0.00	0.00	0.04	0.07	0.11	0.19	0.32	0.37
Other top officers	0.00	0.00	0.00	0.03	0.05	0.10	0.17	0.27	0.31

Total compensation is the sum of salaries, bonuses, long-term bonus payments, and the Black-Scholes value of stock option grants. Long-term pay is payouts from long-term bonus awards made in prior years, including restricted stock. Based on the three highest-paid officers in the largest fifty firms in 1940, 1960, and 1990. In firms where the title "CEO" is not used, the CEO is identified as the president of the company. Other top officers include any executive among the three highest-paid who is not the CEO. The within-firm ratio is the median across firms of the ratio of the CEO's total compensation to the average compensation of the two other highest-paid officers in the firm.

Table 3: compensation by percentile and CEO status

5.4 Figures 4–5 and Table 4: compensation and firm size

Read:

- Figure 4 shows that executives in the largest firms are paid more, but the growth pattern differs by firm-rank group.
- Figure 5 compares median compensation with the S&P index. The two series move closely after 1980 but not in the postwar decades before 1975.
- Table 4 shows that aggregate firm size is strongly related to compensation in 1976–2005 but much less so in 1936–1975 and 1946–1975.
- Firm-specific size remains related to pay, but the market-wide size effect is historically unstable.

Economic message: The modern link between pay and firm size is not a timeless fact. Any theory based on firm size must explain why pay did not rise with firm size from the late 1940s to the 1970s.

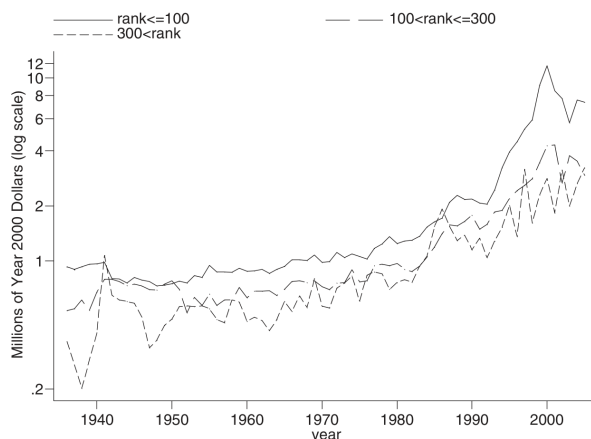


Figure 4
Median total compensation by firm size
Total compensation is composed of salary, bonuses, long-term bonus payments, and stock option grants. Based on the three highest-paid officers in the largest fifty firms in 1940, 1960, and 1990 (a total of 101 firms). Rank ≤ 100 is the median total compensation across all executives in the sample of firms ranked among the top one hundred according to market value in the year; $100 < \text{rank} \leq 300$ is median total compensation for the executives in firms ranked 101 to three hundred; $300 < \text{rank}$ is median total compensation for the executives in all firms

Figure 4: compensation by firm size

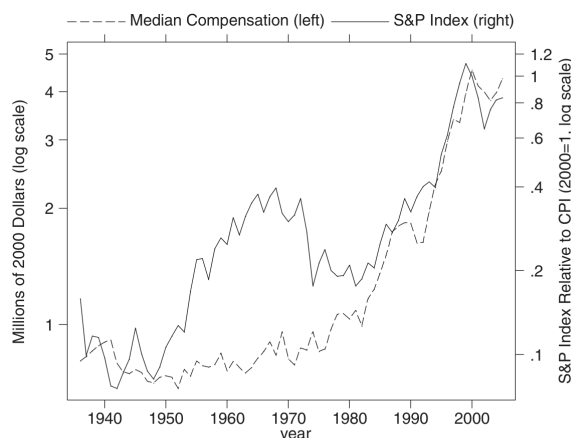


Figure 5
Total compensation and the S&P index
Total compensation is composed of salary, bonuses, long-term bonus payments, and stock option grants. Based on the three highest-paid officers in the largest fifty firms in 1940, 1960, and 1990 (a total of 101 firms). S&P index is expressed relative to the Consumer Price Index and equals 1 in 2000.

Figure 5: compensation and S&P index

Table 4
The correlation between compensation and firm size

	Firm Size = Ln(Market Value)			Firm Size = Ln(Sales)	
	1936–1975 (1)	1946–1975 (2)	1976–2005 (3)	1946–1975 (4)	1976–2005 (5)
Panel 1: DV = $\text{Ln}(\text{Comp}_{it})$					
Average Size in Year $t - 1$	.088** (.026) [.010]	.137** (.025) [.020]	.935** (.035) [.332]	.157** (.032) [.017]	2.65** (0.11) [.259]
Average Firm Size	.208** (.033) [.145]	.212** (.032) [.164]	.292** (.032) [.135]	.305** (.037) [.220]	.358** (.041) [.113]
Size – Firm Avg. – Year Avg.	.182** (.038) [.039]	.199** (.041) [.036]	.264** (.032) [.043]	.240** (.052) [.041]	.346** (.048) [.032]
Panel 2: With Firm Fixed Effects					
Average Size in Year $t - 1$		.135** (.024)	.969** (.037)	.149** (.031)	2.63** (0.11)
Size – Year Avg.		.218** (.040)	.313** (.028)	.277** (.046)	.389** (.046)
Panel 3: Including Lagged Size and Firm FE					
Average Size in Year $t - 1$		–.019 (.037)	.622** (.086)	.240** (.074)	2.31** (0.23)
Average Size in Year $t - 2$		.152** (.041)	.378** (.083)	–.088 (.063)	.362* (.210)
(Size – Year Avg.) in Year $t - 1$		.183** (.040)	.369** (.038)	.203** (.045)	.401** (.067)
(Size – Year Avg.) in Year $t - 2$		.020 (.041)	–.064** (.029)	.084** (.030)	–.040 (.043)
Panel 4: Including Quadratic Time Trend and Firm FE					
Average Size in Year $t - 1$		.035 (.032)	.750** (.082)	.143* (.079)	–.312* (.171)
Size – Year Avg.		.217** (.038)	.309** (.028)	.272** (.046)	.391** (.042)
Panel 5: DV = $\Delta \text{Ln}(\text{Comp}_{it})$					
Δ Average Size in Year $t - 1$		.012 (.031)	.210** (.080)	.087* (.049)	.172 (.174)
Δ Size – Δ Year Avg.		.092** (.032)	.277** (.035)	.077** (.024)	.145 (.116)

The dependent variable in Panels 1 to 4 is $\text{Ln}(\text{total compensation})$ of each executive, and the dependent variable in Panel 5 is the annual change in $\text{Ln}(\text{compensation})$ of each executive. Total compensation is the sum of salaries, bonuses, long-term bonus payments, and the Black–Scholes value of stock option grants, measured in year 2000 dollars and based on the three highest-paid officers in the largest fifty firms in 1940, 1960, and 1990 (a total of 101 firms). Size is measured by the logarithm of the firm's market value in columns (1) to (3) and by the logarithm of the firm's total sales in columns (4) to (5), both measured in year 2000 dollars. Standard errors are shown in parentheses and are clustered by firm. Values in brackets show the fraction of the total variance explained by each independent variable based on an ANOVA decomposition for each complete period. * denotes

5.5 Tables 5–6 and Figure 6: stock holdings and pay-to-performance

Read:

- Table 5 shows that managerial stock holdings relative to shares outstanding fall after the early decades.
- Table 6 shows that the dollar gains from stock and option holdings increase sharply after 1980.
- Combining stock and option holdings, the median executive gains \$120,342 from a 1% return increase in 1990–1999 and \$227,881 in 2000–2005.
- Figure 6 shows that size-adjusted pay-to-performance weakens around the 1970s but rises strongly after the 1980s.

Economic message: Executives were not simply paid like bureaucrats throughout the twentieth century. The strength of wealth-performance incentives was meaningful for much of the sample and became much stronger in recent decades.

Table 5
Managerial stock holdings relative to shares outstanding (percentage points)

	25th percentile	50th percentile	75th percentile	CEOs (median)	Other highest paid officers (median)
1936–1940	.019	.111	.402	.130	.088
1941–1949	.010	.038	.159	.048	.031
1950–1959	.011	.035	.109	.043	.031
1960–1969	.012	.037	.103	.047	.034
1970–1979	.008	.023	.064	.039	.019
1980–1989	.008	.019	.054	.029	.015
1990–1999	.011	.030	.084	.069	.021
2000–2005	.010	.028	.074	.072	.018

The proportion of managerial stock holdings relative to shares outstanding at the twenty-fifth, fiftieth, and seventy-fifth percentiles are based on the three highest-paid executives in the fifty largest firms in 1940, 1960, and 1990 (a total of 101 firms). In firms where the title “CEO” is not used in the proxy statement, the CEO is identified as the president of the company. Other top officers include any executive among the three highest-paid who is not the CEO.

Table 5: managerial stock holdings

Table 6
Ex ante changes in the value of executives’ stock and stock option holdings (year 2000 dollars)

	Option holdings		Stock holdings		Stock + option holdings	
	For \$1,000 change in firm market value (Jensen–Murphy) (1)	For a 1% increase in firm’s rate of return (equity at stake) (2)	For \$1,000 change in firm market value (Jensen–Murphy) (3)	For a 1% increase in firm’s rate of return (equity at stake) (4)	For \$1,000 change in firm market value (Jensen–Murphy) (5)	For a 1% increase in firm’s rate of return (equity at stake) (6)
1936–1940	0	0	1.35	18,401	1.35	18,670
1941–1949	0	0	0.39	6,530	0.40	6,814
1950–1959	0	0	0.31	9,392	0.45	13,975
1960–1969	0.11	7,913	0.35	20,531	0.68	38,978
1970–1979	0.12	6,303	0.22	11,766	0.47	21,743
1980–1989	0.24	13,056	0.17	12,735	0.55	34,679
1990–1999	0.41	57,975	0.29	36,273	0.95	120,342
2000–2005	0.65	127,195	0.27	49,729	1.08	227,881

Based on the three highest-paid executives in the fifty largest firms in 1940, 1960, and 1990. Each column shows the median across all executives in each decade. Ex ante revaluations of stock options are computed using the “delta” for the portfolio held by each executive (Core and Guay 2002). Columns (5) and (6) report the median across executives of the sum of the ex ante revaluations of stock and stock option holdings.

Table 6: ex ante changes in wealth from stock and option holdings

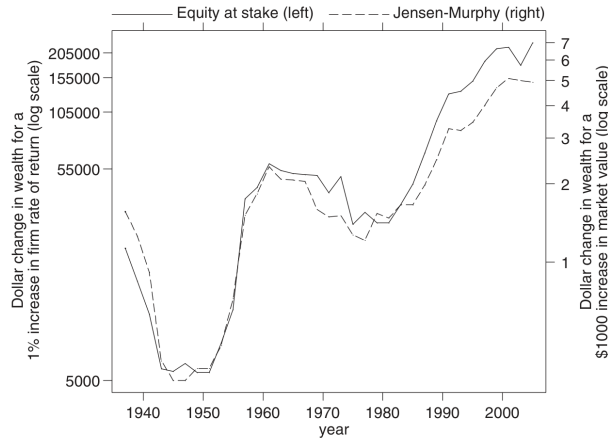


Figure 6
Size-adjusted indexes of pay-to-performance
Based on the three highest-paid executives in the fifty largest firms in 1940, 1960, and 1990 (a total of 101 firms). Results are based on regressing the Jensen–Murphy statistic or value of equity at stake of each individual (which are computed from the ex ante revaluations of stock and option holdings) on indicator variables for time and firm size in a series of overlapping four-year sample periods. Two-year growth rates in pay-to-performance are calculated as the change in the fitted values of these regressions from the first half to the second half of the four-year sample period. The 1937 index values are equal to median pay-to-performance in 1936–1937, and

5.6 Tables 7–8: determinants and strength of incentives

Read:

- Table 7 relates pay-to-performance to governance, board structure, firm size, leverage, growth opportunities, volatility, returns, and industries.
- The determinants of pay-to-performance do not change dramatically across eras.
- Table 8 shows that moving a firm from the median return to the seventieth percentile return creates large gains in executive wealth.
- The dollar change rises from \$265,270 in 1936–1940 to over \$3.2 million in 2000–2005.
- The elasticity of compensation to performance is often large, reaching 4.12 in 2000–2005.

Economic message: Incentives were economically meaningful long before modern option-heavy pay, but the dollar stakes and elasticity grew substantially in recent decades.

Table 7
Correlation of pay-to-performance with firm characteristics

	1936–39	1950–59	1970–79	1990–99
Fraction of inside directors	3.041 (0.990)*	1.002 (0.677)	0.038 (0.934)	1.138 (1.042)
Ln(# directors)	1.584 (0.572)*	0.082 (0.366)	−0.424 (0.483)	−0.862 (0.518)
Ln(average market value in past 3 years)	0.323 (0.210)	0.319 (0.125)*	0.405 (0.102)*	0.631 (0.106)*
Leverage _{<i>t</i>−1}	−0.655 (1.087)	3.453 (0.754)*	2.067 (1.201)	0.499 (0.482)
Growth opportunities _{<i>t</i>−1}	0.464 (0.155)*	0.711 (0.308)*	0.613 (0.199)*	0.188 (0.083)*
Idiosyncratic stock-price volatility _{<i>t</i>−1}	0.087 (0.498)	−0.474 (0.421)	1.052 (0.378)*	0.415 (0.280)
Real rate of return _{<i>t</i>−1}	−0.169 (0.072)*	0.088 (0.201)	0.053 (0.116)	0.127 (0.132)
Industry indicators (relative to other manufacturing)				
Railroad	−2.332 (1.164)*	−0.336 (0.369)	−1.152 (0.321)*	0.904 (0.521)
Communications	−	−1.929 (0.399)*	−2.047 (0.403)*	−1.769 (0.419)*
Utilities	−1.040 (0.573)	−2.187 (0.439)*	−1.653 (0.543)*	−2.055 (0.521)*
Trade	0.547 (0.453)	−0.327 (0.422)	0.264 (0.263)	−0.057 (0.238)
Finance, insurance, and real estate	0.167 (0.750)	−1.025 (0.501)*	−0.438 (0.551)	0.322 (0.348)
Chemical manufacturing	0.204 (0.613)	0.201 (0.405)	0.121 (0.359)	0.335 (0.178)
Industrial equipment	0.929 (0.702)	−0.377 (0.551)	0.334 (0.393)	−0.296 (0.284)
Electronic equipment	−2.354 (0.608)*	−0.401 (0.280)	−0.091 (0.434)	0.409 (0.472)
Number of observations	404	1432	2089	1814
Adjusted R-squared	0.37	0.43	0.36	0.52

Pay-to-performance is defined as $\ln(\text{equity at stake})$, but results are similar when using $\ln(\text{Jensen-Murphy statistic})$. Fraction of inside directors is the number of directors that are managers of the firm relative to the total number of directors. Leverage is the ratio of total liabilities to assets. We proxy for growth opportunities with the market-to-book ratio, calculated as sum of total liabilities and market value of equity divided by assets. Idiosyncratic stock-price volatility is the five-year standard deviation of the residuals from a regression of the firm's rate of return on the rate of return of the S&P 500. The omitted industry indicator corresponds to all other manufacturing firms in our sample. Regressions also include year and job title indicators. Standard errors are clustered by firm. * denotes statistical significance at the 5% level.

Table 7: *pay-to-performance and firm characteristics*

Table 8
The strength of managerial incentives: change in wealth due to raising the firm's rate of return from the fiftieth to the seventieth percentile

\$ change	Median across executives		
	Percentage change in wealth =		Elasticity =
	(1)	(2)	(3)
in wealth	total comp. + Δ wealth at rate of return ^{50th}	rate of return ^{70th} − rate of return ^{50th}	
(1)	(2)	(3)	
1936–1940	265,270	28.6	2.00
1941–1949	96,947	9.2	0.64
1950–1959	199,046	22.9	1.60
1960–1969	556,247	50.7	3.54
1970–1979	312,559	29.0	2.03
1980–1989	496,266	27.6	1.92
1990–1999	1,720,953	52.3	3.66
2000–2005	3,212,822	59.0	4.12

Note: Based on the three highest-paid executives in the fifty largest firms in 1940, 1960, and 1990 (a total of 101). The dollar change in wealth is defined as the ex ante revaluation of stock and option holdings assuming a rate of return at the seventieth percentile (22.7%) minus the revaluation evaluated at the fiftieth percentile rate of return (8.4%). Column (1) shows the median of this value across executives in each decade. The percentage change in wealth is the dollar change in wealth for each individual executive divided by the total change in an executive's wealth at median firm performance, defined as the revaluation of stock and option holdings at the fiftieth percentile rate of return plus salaries, bonuses, and stock option grants. Column (2) presents the median of this value across executives in each decade. Column (3) approximates an elasticity of changes in wealth to changes in firm performance by dividing column (2) by the percentage difference in firm value between the fiftieth and seventieth percentiles of firm performance ($22.7 - 8.4 = 14.3\%$).

Table 8: *strength of managerial incentives*

6 Short summary

- Frydman and Saks build a long-run dataset on top executive pay from 1936 to 2005.
- Median real compensation was surprisingly flat from the late 1940s to the mid-1970s.
- Pay rose sharply after the 1980s, driven partly by stock options and long-term incentive pay.
- The modern correlation between pay and aggregate firm size was weak before the mid-1970s.
- This weak earlier correlation challenges explanations that attribute pay growth solely to firm-size growth or the market for talent.
- Executive wealth was sensitive to firm performance for much of the sample, not only in the modern period.
- Pay-to-performance weakened around the 1970s and then increased strongly after the mid-1980s.
- CEO pay grew faster than pay for other top executives, especially after 1980.
- Pre-tax and after-tax pay follow similar trends, so tax changes do not explain the main pattern.
- The long-run evidence challenges simple governance, risk, tax, and firm-size explanations for the rise in executive pay.

7 Conclusion

7.1 Core conclusion

Frydman and Saks show that the modern rise in executive compensation is historically unusual. For several decades after World War II, executive pay remained nearly flat even though firms grew. The recent surge in pay, especially after the 1980s, is therefore not just the continuation of a stable long-run relationship between firm size and compensation.

7.2 Economic intuition

The paper's central intuition is that executive compensation institutions changed over time. In the earlier decades, large-firm growth did not mechanically translate into higher pay, and equity-linked compensation was more limited. In recent decades, compensation became more tied to firm size, equity markets, and performance-based wealth. These changes suggest that the market for executives, board practices, incentive design, and the nature of managerial skills evolved together.

7.3 Takeaway

The long-run perspective makes the modern executive-pay boom harder to explain with any single theory. Firm size, incentives, taxes, risk, governance, and managerial talent all matter, but each must explain why the postwar decades looked so different from the post-1980 period.